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Department of Computer Science and Engineering

Dhulikhel, Kavre



A Project Report on

"Smart-Beema"

[Code No: COMP 308]

(For partial fulfilment of 3rd Year, 2ndSemester in Computer Engineering)

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Bonafide Certificate

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Here again thanking those beautiful hands and great minds that have helped us to complete this project.

Abstract

Companies are not always looking out for the best interests of clients. Fraud on a daily basis is not rare these days and among the worst culprits are insurance firms and brokers. While there are laws and regulations in place to deal with such frauds, following them is a time-consuming procedure, especially in Nepal, where administrative chores take a long time. Even though there are a variety of insurance solutions available to consumers, the adoption of insurance policies is not so popular due to lack of trust in insurers and the tedious process of getting insured. **Smart-Beema** is a software product that aims to solve difficulties with traditional insurance by allowing consumers to enter into trust-less agreements and simplifying the claim process. It does so via Smart Contracts using Solidity and deploying a user interface built using the React framework that facilitates easy interaction with the Ethereum Blockchain network. Smart-Beema can be used by an insurance company or a government to provide parametric based insurance service to users with automatic claim settlement process as the main feature ultimately improving insurance adoption.

Keywords: *Fraud, Ethereum Blockchain, Smart Contract, Solidity, React*

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Abbreviations

AI: Artificial Intelligence

P2P: Peer to Peer

UI: User Interface

API: Application Programming Interface

IDE: Integrated Development Environment

JSX: JavaScript XML

DOM: Document Object Model

CDN: Content Delivery Network

HTML: HyperText Markup Language

CSS: Cascading Style Sheets

SPI: Standardised Precipitation Index

ML: Machine Learning

IOT: Internet Of Things

Chapter 1: Introduction

1.1 Background

The world we live in is driven by self-interest and profit. Companies and individuals are not always looking out for our best interests. Fraud is a typical occurrence in Nepal's underdeveloped economy, and it continues to be a pervasive scourge that affects, among other things, the insurance sector. One of the worst offenders is insurance firms and brokers.

Insurance firms are responsible for reimbursement in the event of a policyholder's loss, but in practice, the corporations often appear to have significant conflicts of interest since they do not want to pay. As a result, they make every effort to locate loopholes in the contract in order to get out of the position. While there are laws and regulations in place to deal with such circumstances, following them is a time-consuming process that can be costly for the policyholder. To reclaim what is properly theirs, they must go through a lengthy process. Many insurance solutions are beyond of reach for smallholder farmers, and there is a lack of faith in insurers as a result of past delays in disbursing compensation of up to three or even six months, causing insurance uptake to lag.

Smart-Beema is a software product that aims to solve difficulties with traditional insurance by requiring consumers to enter into trust-less agreements via smart contracts and facilitates easy interaction with the Ethereum blockchain network. [1] Our project intends to provide smallholder farmers with crop insurance using a novel sort of parametric insurance solution known as weather index insurance. Farmers are covered by this insurance in the event of bad weather that could reduce their production. It provides payments to insured farmers immediately based on a specified index, such as rainfall, as monitored by meteorological stations or satellites, without consideration for actual damage assessment. Because there are no claim assessors involved, this approach is faster, cheaper, and more reliable.

1.2 Objectives

There are some specific goals and objectives for our project. Some of them are listed below:

- To remove counterparty risk through trust minimised agreements.
- To improve user experience by simplifying and speeding up the claim process.

1.3 Motivation and Significance

The Nepal government introduced the Agriculture and Livestock Insurance program in the fiscal year 2013/2014 to cover the damages and losses to agriculture. Only roughly 200,000 farmers have profited from the insurance scheme in a country of about 30 million people, with over 60% of the population reliant on agriculture for a living. There are three main reasons for the low number of agriculture and livestock insurance holders, according to Senior Agriculture Extension Officer Shiva Sundar Ghimire of the Department of Agriculture. [2]

Smart-Beema intends to address this issue by creating the 'Generic Insurance Framework,' a decentralised, digital insurance platform based on the Ethereum blockchain that would digitise parametric insurance products. The full life cycle of an insurance policy may be automated as software code using Smart-Beema, resulting in a so-called "insurance smart contract," which drastically reduces insurance operational costs while enhancing transaction speed and transparency. It focuses almost solely on weather index insurance, a relatively new sort of parametric insurance solution. Farmers can get weather index insurance to assist them manage the risks that come with bad weather. Rather than paying benefits based on an assessment of the actual damage, it pays payments parametrically based on a specified index such as rainfall, as detected by weather stations or satellites. As a result, they don't need claims assessors and can settle claims more cheaply, quickly, and objectively.

Chapter 2: Related Works

Blockchain technology is fast gaining traction in the insurance business due to its ambitious potential to drive simplicity and efficiency through the construction of new financial services infrastructure and processes. Blockchain and smart contracts often combined with other emerging technologies such as Internet of Things (IoT) and Artificial Intelligence (AI), can be used for managing claims in a responsive and transparent way, lowering administration and underwriting costs and supporting more accurate pricing. Automated claims submission and processing could improve claims assessment, fraud detection, and ensure more timely payments.

2.1. Etherisc

Etherisc is an open-source development platform that focuses on decentralised insurance applications. Blockchain insurance application: Etherisc builds decentralised, blockchain-centric applications for different sectors of the insurance industry. The company is focused on using ledger technology to cut down on inefficiencies, namely high processing fees and extensive claim-processing times. [3]

Etherisc has already developed six different decentralised insurance-related applications. One of them is a crop insurance app with which farmers identify their land and crops as well as any losses due to weather. Another app insures Etherisc members against potential crypto wallet hackings.

2.2. Lemonade

Lemonade combines AI and Distributed Ledger Technology to offer insurance to renters and homeowners starting at $5 and $25 a monthly, respectively.

At Lemonade, blockchain comes into play through smart contracts. The company's business model takes a fixed fee from each monthly payment and allocates the rest towards future claims. If a claim is made, the blockchain's smart contracts will immediately attempt to verify the loss so a customer can get paid quickly. If a claim is approved, Lemonade's AI and

blockchain combination would pay the customers in three seconds. (Lemonade was voted #1 of 270 companies for customer satisfaction in renter's insurance.) [4]

2.3. Fizzy

Fizzy, a flight delay insurance tool, is a subsidiary of global insurance giant AXA.Fizzy uses blockchain to ensure that members whose flights are delayed more than two hours are immediately compensated. The company's blockchain supplements travel insurance that usually doesn't cover financial loss due to flight delays.

The tool uses smart contracts to lock in terms for payments and policy information. Users have only to enter their flight details, personalise their coverage and make a payment. Fizzy will then use blockchain to immutably verify flight delay data and compensate customers. By December 2018, Fizzy's instant blockchain-based flight delay payments covered 80% of all worldwide flights. [5]

Chapter 3: Design and Implementation

Blockchain uses ‘blocks’ of information to keep track of data transactions in a distributed network of multiple nodes or computers. A transaction with party B is requested by party A, such as transferring money, setting up a contract, or sharing records. This transaction is broadcasted to a distributed network of ‘nodes’ or computers which will validate it according to an agreed set of rules called ‘consensus’ mechanism. When the transaction is validated, a new ‘block’ will be added to the blockchain. When a new block is added to the blockchain, it is timestamped, a pointer to the previous block in the chain is provided, and the transaction data entered. After that it is processed by the cryptographic technique of hashing where a hash is calculated on the hash of the previous block plus the data contents of the new block. The result then becomes the hash of the new block. This process ensures that each block is linked to the previous one, thereby forming a chain of blocks. The unique record that forms a blockchain is shared by each node or computer in the network and is constantly updated and synchronised. As a database or ledger, blockchain ultimately stores the records of all transactions executed across the network.

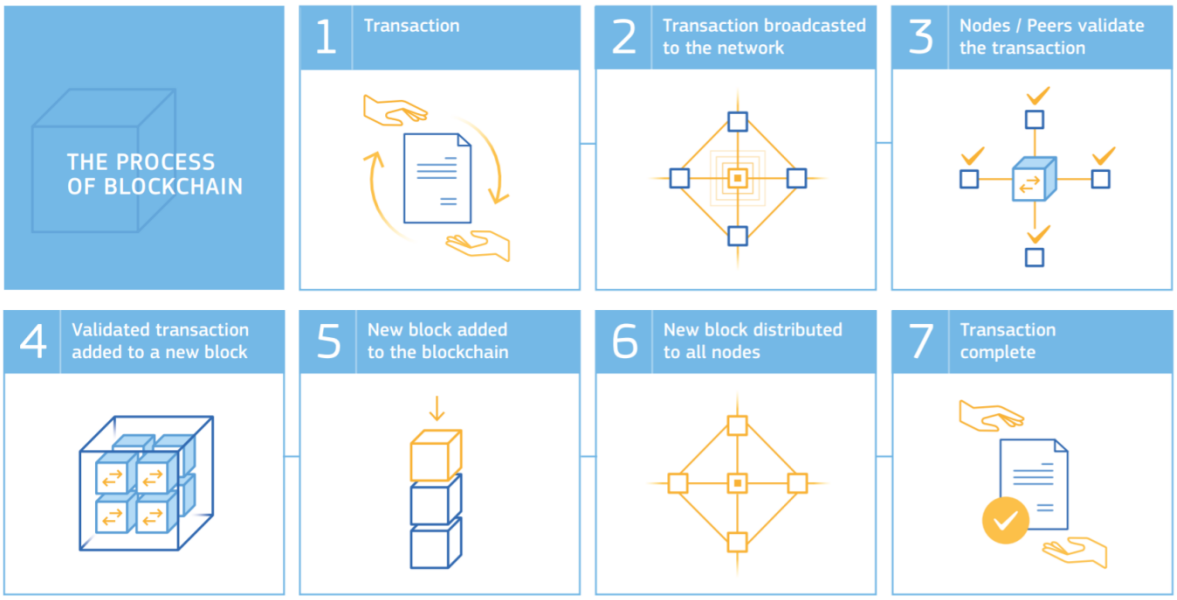


Figure 3.1: The process of blockchain

When the smart contract is deployed on the blockchain, they are stored on a public database and cannot be changed. The transactions that happen in a smart contract are processed by the blockchain, which means they can be sent automatically without a third party.

The user registers for the insurance with a premium amount paid through the metamask wallet and when the time comes for the user to claim payout, the claim process is automatically handled without any human interruption. [6]

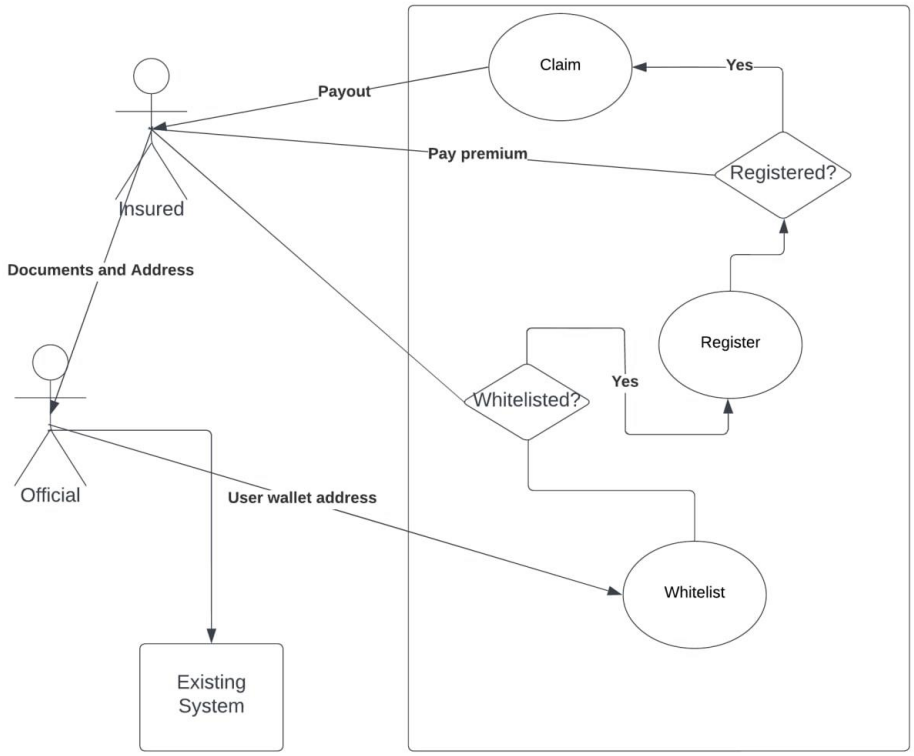


Figure 3.2: Use case diagram for Smart-Beema

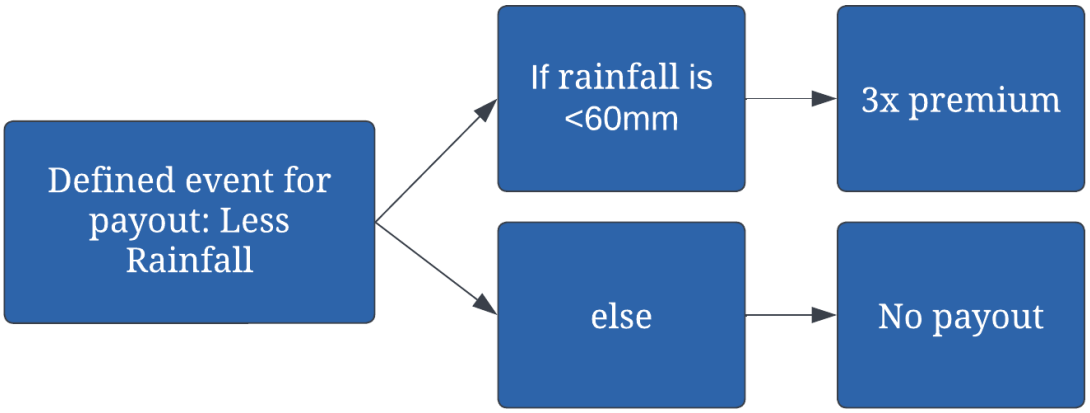


Figure 3.3: Simplified smart contract logic

3.1 Research and Study

The blockchain technology is still in its infancy, with many of its potential applications yet to be discovered. We used the knowledge, information, and experiences gained from researching previous products and projects that used similar ideas, as well as read numerous web guides and videos, to produce a viable project of our own. To create and deploy our system, we used Ethereum Blockchain technology on the rinkebytestnet, as well as the Standard Precipitation Index (SPI), which is based on the probability of precipitation for any time scale and may be used to detect and characterize meteorological droughts. [7]

3.2 Design, Wireframing and Prototyping

We used Figma which is a vector graphics editor and prototyping tool. Since it is a web-based application with additional offline features enabled by desktop applications for macOS and Windows, we used it as it emphasises real-time collaboration. It is also starting to become a great choice especially for beginners, due to its straightforward and compact design and availability.

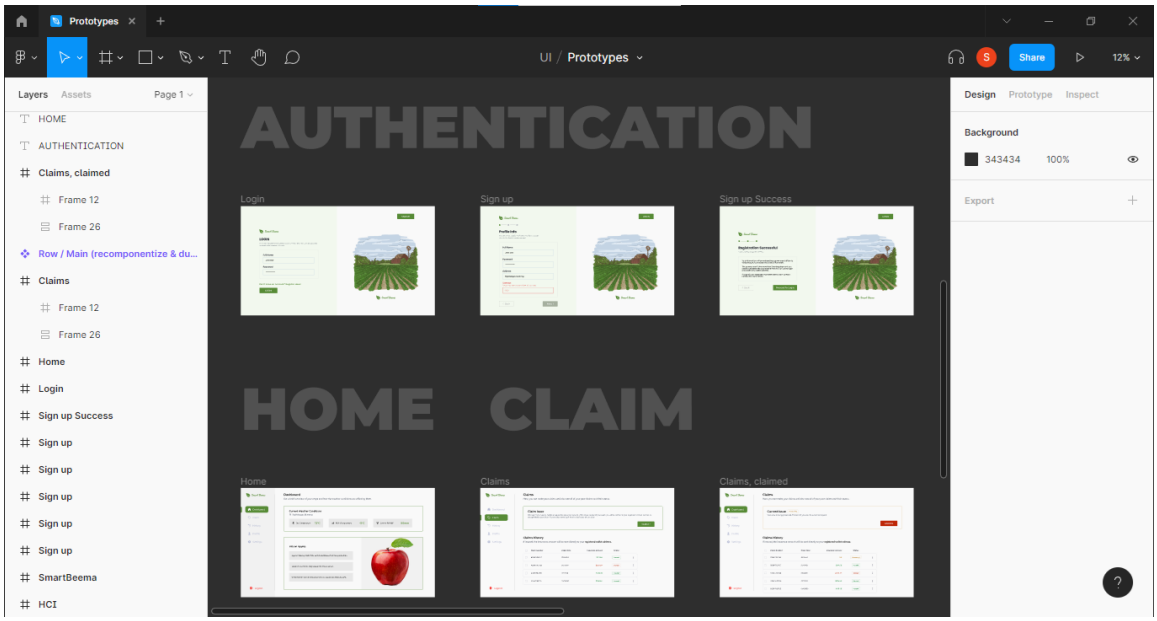


Figure 3.4: Design prototypes made in Figma

3.3 Core Programming

We built our application using Solidity to write the smart contracts, Remix to deploy the contracts to the testnet, and React to build the front end.

3.3.1 Client-Side (Frontend)

Front-end involves the codebase that runs on the client computer.

React

React is a JavaScript library for building user interfaces. React was used as it is a declarative and component-based javascript library that makes it easy and efficient to create interactive UIs. It was used as the reusable components along with simple JSX syntax makes it straightforward and coherent to use it in the front-end development. Also, the virtual DOM property of React made it easy for us to design, develop and code in this JavaScript library.

React-Bootstrap

React-Bootstrap is a component-based library that provides native Bootstrap components as pure React components. We used it for the elements and components in the React app (like Container, Form, Row, Column, Nav, etc.). Instead of utilising JavaScript source and plugins from the CDN, it converts all JavaScript to React and bundles all components together.

JSX

JSX is simply a syntax extension of JavaScript. JSX was used as it allowed us to directly write HTML in React (within JavaScript code). We used it to create components and templates in React as it comes with the full power of JavaScript. It is also faster than normal JavaScript as it performs optimizations while translating to regular JavaScript.

CSS

Cascading Style Sheets is a stylesheet language used to describe the presentation of a document written in HTML or XML. CSS was used as it describes how elements should be rendered on screen, or on other media. The styling of the pages and their components like texts, icons, buttons, input fields, sidebar, etc. were done easily with CSS.

3.3.2 Backend

The “back-end” or the smart contract of our project was written in solidity in the remix IDE and deployed to the testnet using remix IDE. Metamask wallet was used in order to connect to the Kovan testnet which provides access to the Ethereum blockchain. We used the “faucets” available on the web to acquire Kovan Ether without the need for mining.

MetaMask

Metamask is a software cryptocurrency wallet used to interact with the Ethereum blockchain. Since Metamask allows users to access their Ethereum wallet and the test networks, we used it to interact with our decentralised application. Once the user logs in to their metamask wallet they can interact with our application.

Solidity

Solidity is a statically typed curly-braces programming language for creating Ethereum-based smart contracts. Smart contracts are programs that are performed within a peer-to-peer network with no central authority, allowing for the implementation of tokens of value, ownership, voting, and other types of logic. Using solidity we wrote our smart contract and deployed it on the kovan testnet using the metamask wallet.

Remix

Remix IDE is an open source web and desktop program which provides a framework for plugin-based development tools.. With remix’s graphical interfaces for functions and being able to write contracts easily on the browser, it was much easier to debug and interact with our smart contract. By directly connecting remix with our metamask wallet it made writing and testing our smart contract easier and faster. By deploying the contract on testnet we made sure that the contract would perform just as well on the mainnet if need be.

3.4 System Requirements

3.4.1 Requirement Specification

Smart-Beema is built to address the issues that arise due to conflicts of interest between the insurer and the insured. Thus, it should be able to facilitate trust-minimised agreements between the two parties who are the primary actors of the system.

Smart-Beema is designed to fit alongside any existing systems that the insurer party runs and maintains. Inclusion of a wallet address alongside the other data they hold, gives them the ability to utilise Smart-Beema as a platform. They can use this platform and charge their users the insurance premium fee and automate the claim process, thereby reducing the operating cost.

Taking these factors into consideration, the functional requirement of the project is to allow for users of the insurer to be able to select an insurance plan, pay premium through their crypto wallet, and initiate a claim based on the terms of the insurance policy.

Furthermore, some non-functional requirements of the project are as follows:

- Ease of use
- Performance
- Reliability
- Compatibility

3.4. Hardware and Software Requirements

Any web browser with metamask wallet extension installed with support for HTML5, CSS and JavaScript is capable of running our web app. We recommend Mozilla Firefox or Google Chrome.

Chapter 4: Discussion on Achievements

Although the rise of blockchain technology has generated massive interest among governments, enterprises, and academics, because of its capability of providing a transparent, secured, tamper-proof solution for interconnecting different stakeholders in a trustless setup, implementing blockchain technology to create a product was equally difficult. [8] Because of the lack of resources on blockchain and smart contracts, the implementation process turned out to be quite challenging. We deviated from our initial plan of using the SPI (Standardised Precipitation Index) to use it as a rain parameter and ended up using the accumulated rainfall over a period of time. Getting the accumulated historic rainfall data was one of the challenges we faced as using such services inside the blockchain would require us to pay the gas fees. We also had to implement an approach different from our original intent which was for the product to work completely online. The verification process of the documents was not possible to perform online so we went to an offline model of verification where an official would perform the verification.

Despite all the challenges faced during our work, at the end of the project we became familiar with blockchain technology, how it works, its applications, and how smart contracts can be used for insurance and other financial purposes like trading, investing, lending, and borrowing.

Features

The notable features of our project are as follow:

- Simple and easy to use user interface.
- Automatic claim settlement.
- Reduced time interval for claim settlement.
- Simple to integrate with existing record keeping systems for the insurer.

Chapter 5: Conclusion

Blockchain has the potential to be utilised throughout the whole insurance value chain. It has the ability to provide major digital opportunities, decrease process duplication, increase process automation, assist in cost reduction, increase efficiency, improve customer experiences, and improve data quality, collecting, and analytics. It could also allow for the creation of new products and services, as well as the adoption of insurance platforms and ecosystems, improved interaction with third parties, the promotion of completely decentralised peer-to-peer (P2P) insurance business models, and the implementation of parametric insurance products.

Our system is a decentralised, digital, parametric insurance platform based on the Ethereum blockchain. The entire life cycle of an insurance policy can be automated as software code, a so-called 'insurance smart contract' dramatically reducing operational costs of insurance while increasing the speed of transactions and transparency. They therefore require no claims assessors and allow for cheaper, faster and more objective claims settlement.

We feel that we have done justice to what our intended envision for how Smart Contracts can be deployed to address real world issues using the available tools and knowledge accumulated in a short period of time. However, we still have not been able to harness the full potential of the project as we fell short of time as well as the knowledge and experience. Positively, our inexperience is an indicator and a motivation to further learn and accumulate experience to improve the project and ultimately ourselves.

5.1 Limitations

Following are some of the limitations of our project in its current state which could not be addressed either due to our lack of knowledge or factors that are beyond our control.

- Users have to use a third party wallet to use the service.
- Users can insure for one type of crop only.
- Bitcoin and blockchain technology is currently banned in Nepal.
- Dependent on the weather data provider.
- Parameters may not always model real world scenarios.

5.2 Future Enhancements

Though we were able to implement the core idea behind the project, we still have not been able to bring the project to its full potential. Following are some enhancements that we have planned to add to our project in the days to come.

- Modify the app to include the transaction statements within the app itself.
- Create our own Oracle Nodes for providing weather or other relevant parametric data.
- Possibility of creating our own blockchain and digital currency.
- Add support for other parameter based insurance and the ability to register for more than one type of insurance.
- Use of Artificial Intelligence (AI) and Machine Learning (ML) along with satellite data to create a better parameter model.

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Appendices

Gantt Chart

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Research							
Design							
Front-end							
Smart Contract							
Integration/Testing							
Documentation							

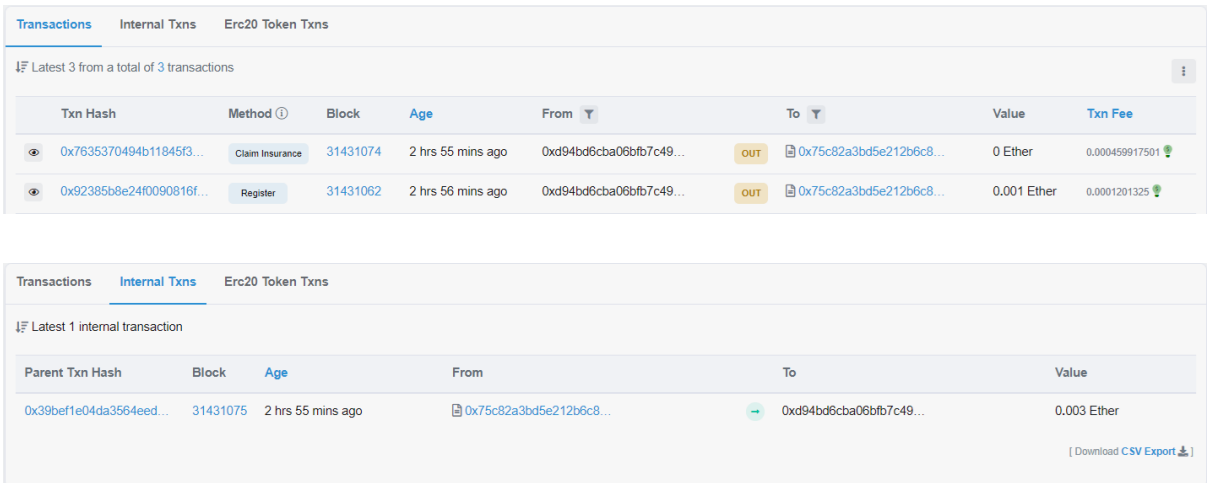
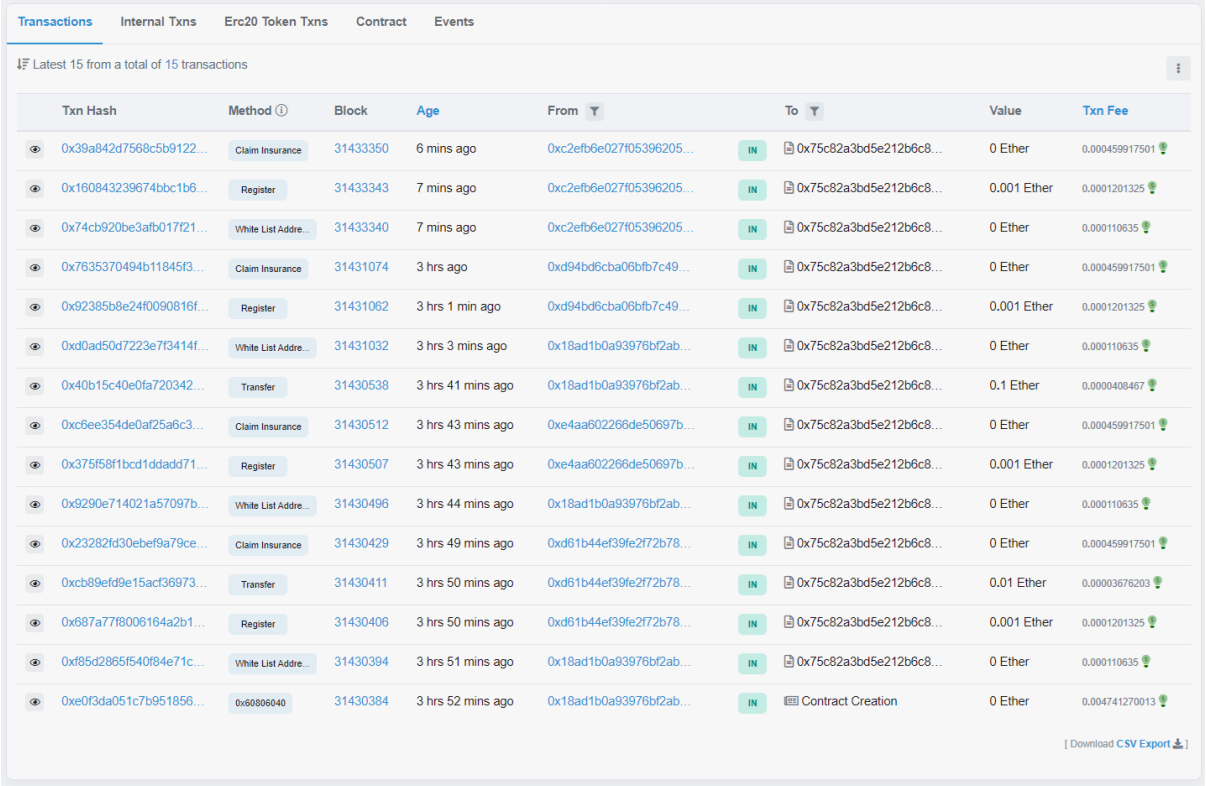


Fig 5.1 Transaction statement details of the user on Kovan Etherscan



Transactions | Internal Txns | Erc20 Token Txns | Contract | Events

Latest 15 from a total of 15 transactions

Txn Hash	Method	Block	Age	From		To	Value	Txn Fee
0x39a842d7568c5b9122...	Claim Insurance	31433350	6 mins ago	0xc2efb6e027f05396205...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000459917501
0x160843239674bbc1b6...	Register	31433343	7 mins ago	0xc2efb6e027f05396205...	IN	0x75c82a3bd5e212b6c8...	0.001 Ether	0.0001201325
0x74cb920be3afb017f21...	White List Addre...	31433340	7 mins ago	0xc2efb6e027f05396205...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000110635
0x7635370494b11845f3...	Claim Insurance	31431074	3 hrs ago	0xd94bd6cba06bfb7c49...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000459917501
0x92385b8e24f0090816f...	Register	31431062	3 hrs 1 min ago	0xd94bd6cba06bfb7c49...	IN	0x75c82a3bd5e212b6c8...	0.001 Ether	0.0001201325
0xd0ad50d7223e7f3414f...	White List Addre...	31431032	3 hrs 3 mins ago	0x18ad1b0a93976bf2ab...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000110635
0x40b15c40e0fa720342...	Transfer	31430538	3 hrs 41 mins ago	0x18ad1b0a93976bf2ab...	IN	0x75c82a3bd5e212b6c8...	0.1 Ether	0.0000408467
0xc6ee354de0af25a6c3...	Claim Insurance	31430512	3 hrs 43 mins ago	0xe4aa602266de50697b...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000459917501
0x375f58f1bcd1ddadd71...	Register	31430507	3 hrs 43 mins ago	0xe4aa602266de50697b...	IN	0x75c82a3bd5e212b6c8...	0.001 Ether	0.0001201325
0x9290e714021a57097b...	White List Addre...	31430496	3 hrs 44 mins ago	0x18ad1b0a93976bf2ab...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000110635
0x23282fd30ebef9a79ce...	Claim Insurance	31430429	3 hrs 49 mins ago	0xd61b44ef39fe2f72b78...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000459917501
0xcb89efd9e15acf36973...	Transfer	31430411	3 hrs 50 mins ago	0xd61b44ef39fe2f72b78...	IN	0x75c82a3bd5e212b6c8...	0.01 Ether	0.00003676203
0x687a77f8006164a2b1...	Register	31430406	3 hrs 50 mins ago	0xd61b44ef39fe2f72b78...	IN	0x75c82a3bd5e212b6c8...	0.001 Ether	0.0001201325
0xf85d2865f540f84e71c...	White List Addre...	31430394	3 hrs 51 mins ago	0x18ad1b0a93976bf2ab...	IN	0x75c82a3bd5e212b6c8...	0 Ether	0.000110635
0xe0f3da051c7b951856...	0x60806040	31430384	3 hrs 52 mins ago	0x18ad1b0a93976bf2ab...	IN	Contract Creation	0 Ether	0.004741270013

[Download CSV Export]

Fig 5.2 Transaction details of the contract on Kovan Etherscan

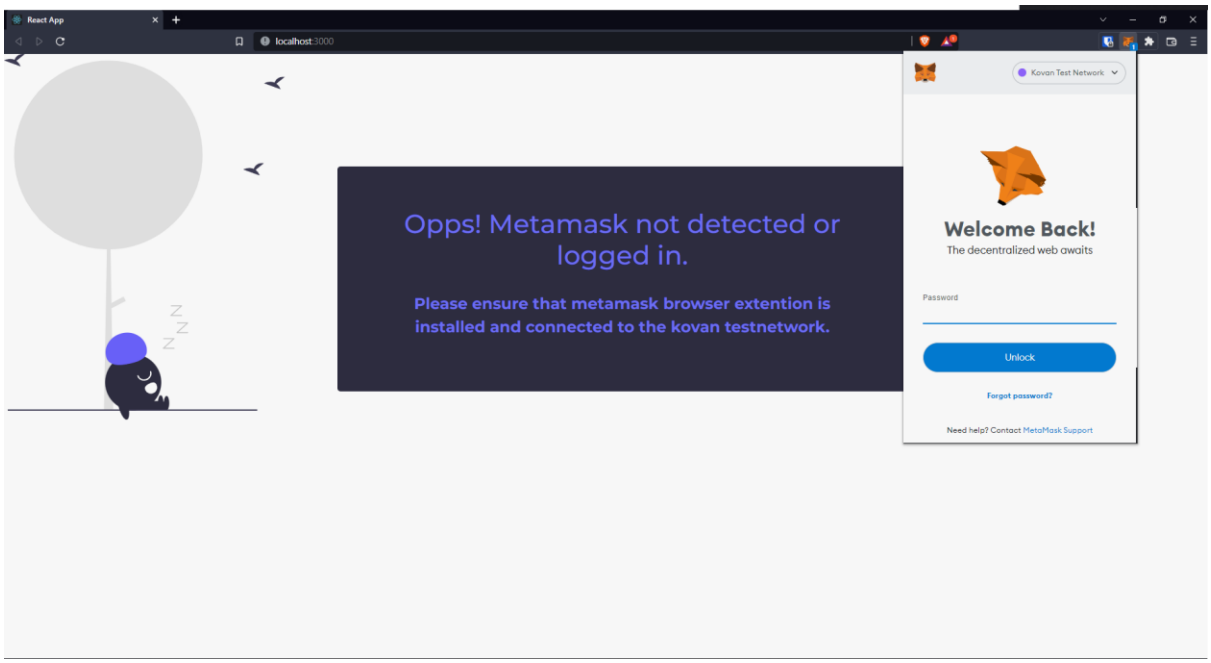


Fig 5.3 Error page when MetaMask is not connected

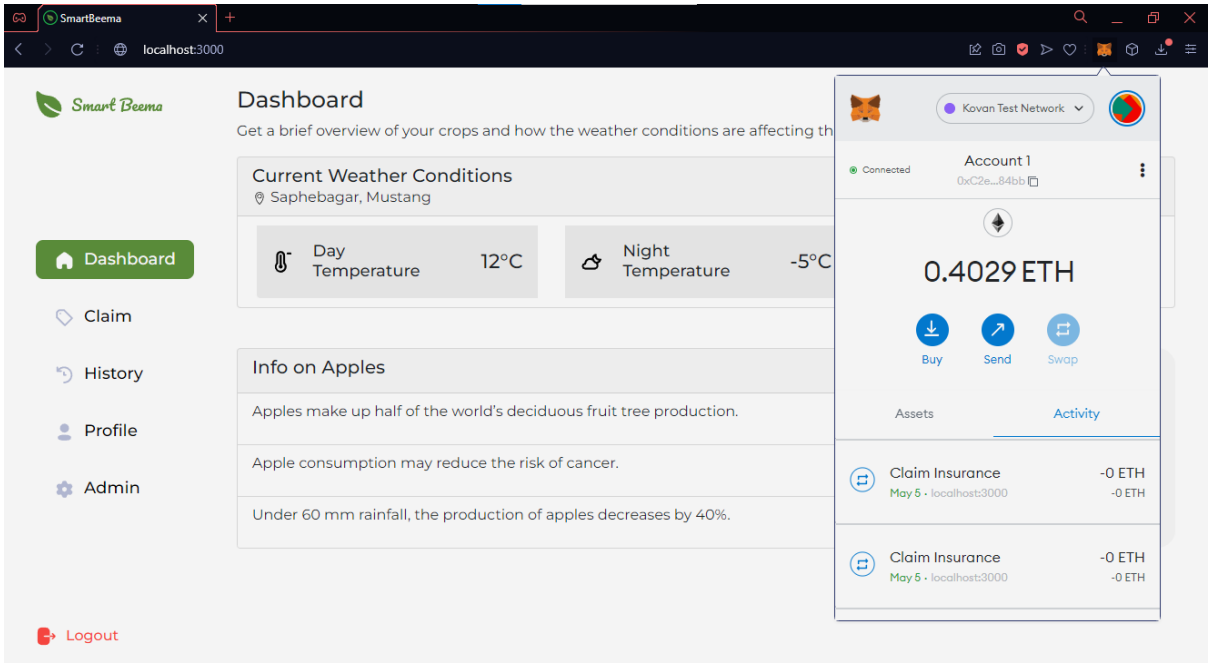


Fig 5.4 Dashboard page and MetaMask popup after logging in

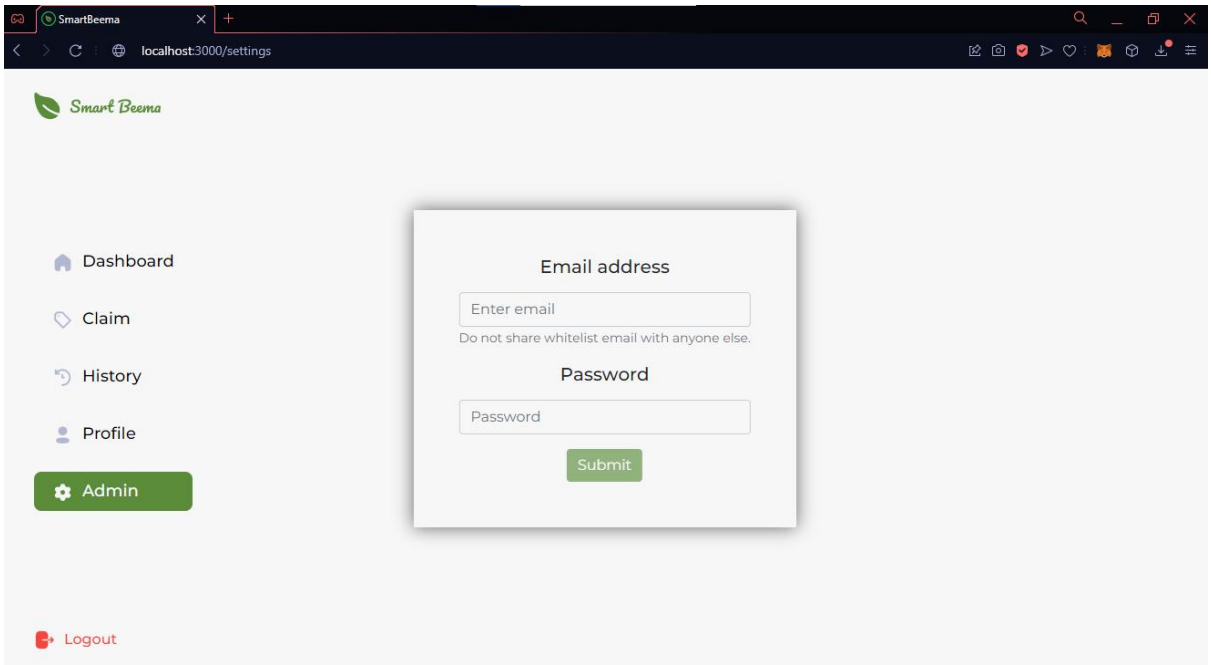


Fig 5.5 Login page for the administrator

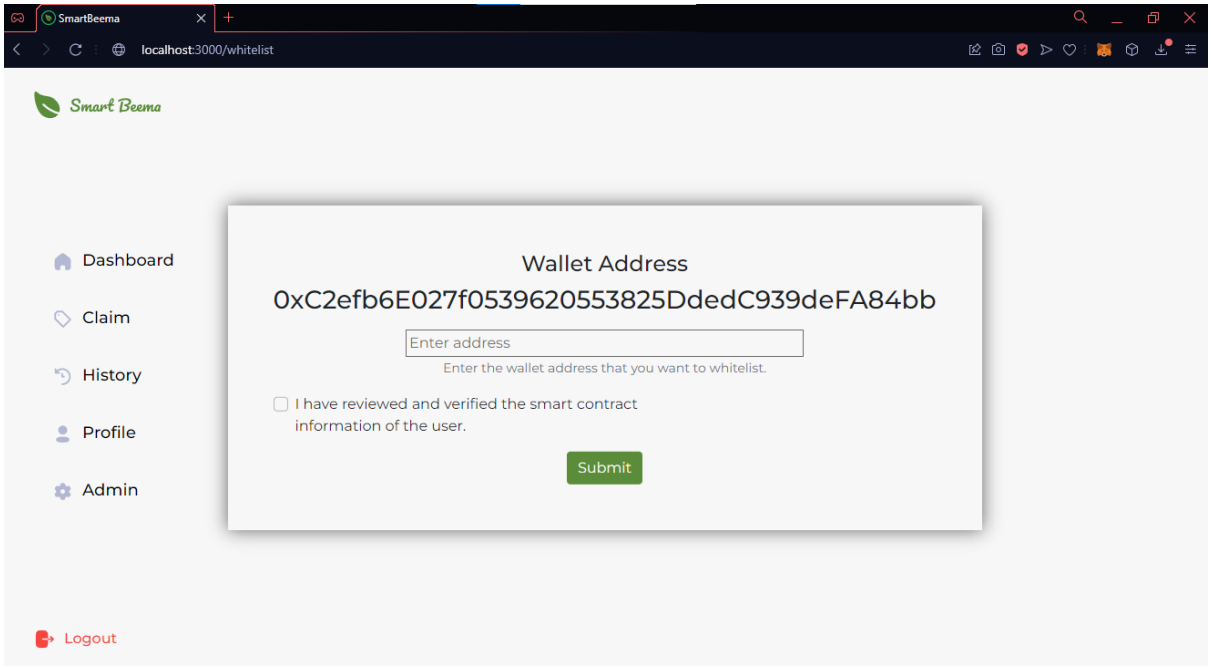


Fig 5.6 Page for Admin to whitelist the wallet address

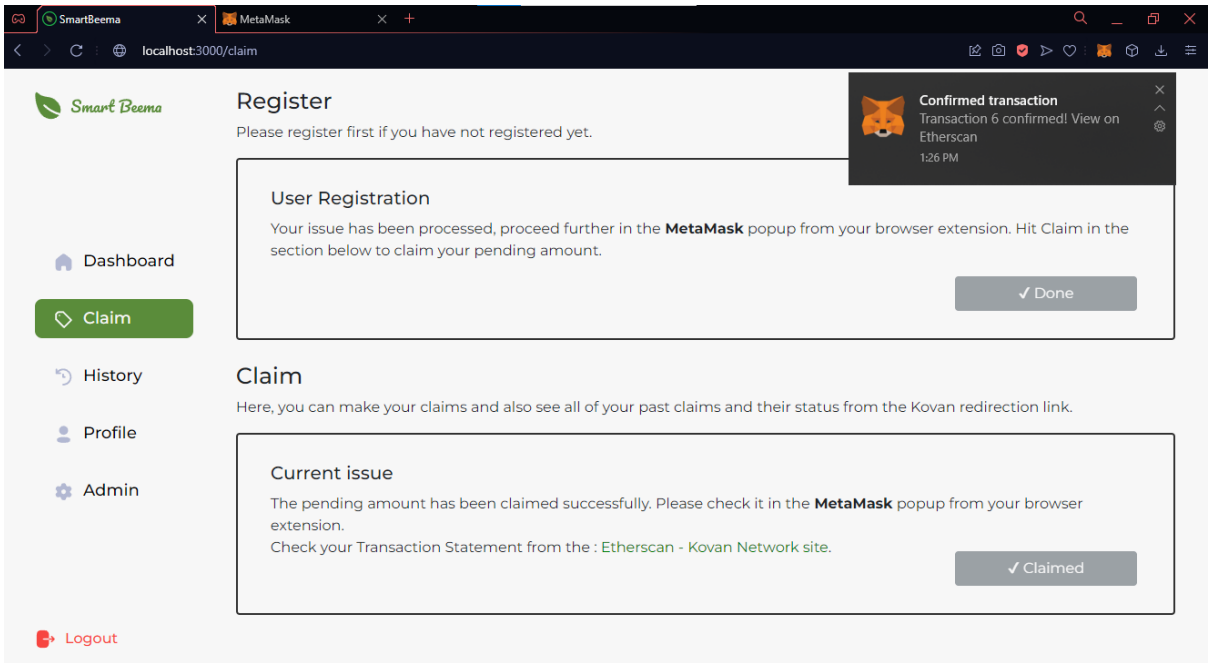


Fig 5.7 Claim and Register page for the user showing confirmed transaction

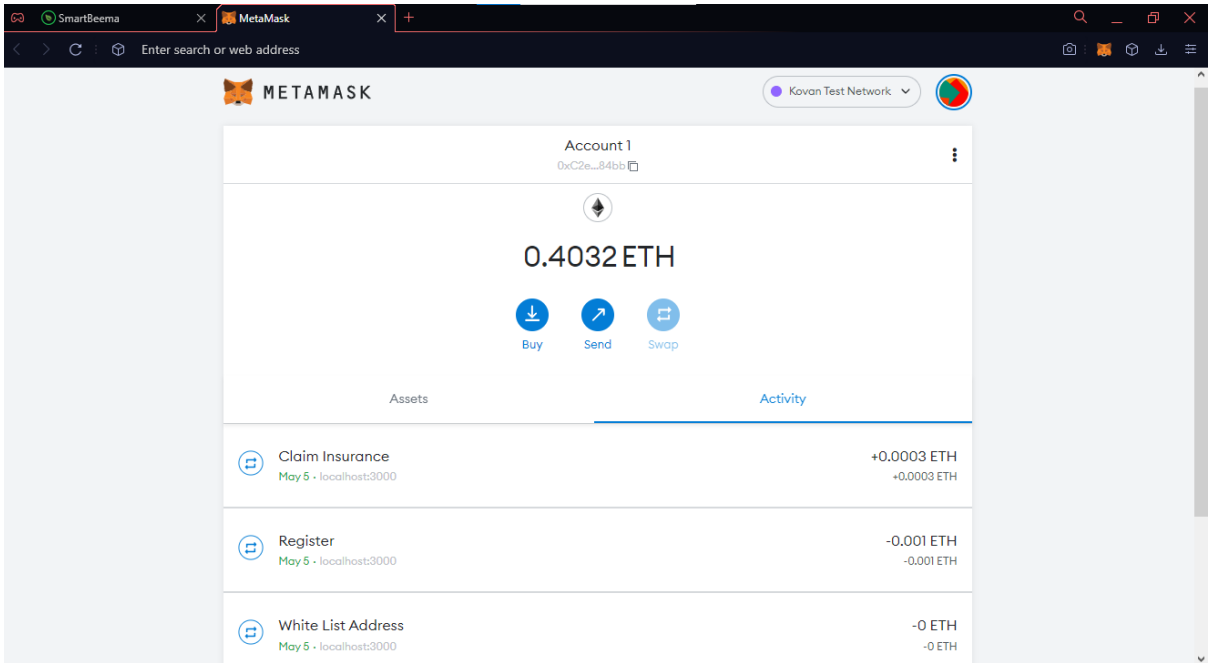


Fig 5.8 MetaMask wallet layout showing recent activity